

Tarun Emmanuel Majhi

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Summary

AI/ML Software Engineer with 3+ years of experience building scalable, production systems. Strong expertise in LLMs, distributed data pipelines, and cloud-native infrastructure, with a proven record of improving model performance and delivering measurable business impact. Experienced in end-to-end system design, from data processing and model development to deployment and optimization using Python, AWS, and Kubernetes.

Skills

Programming Languages & Tools: Python, Java, SQL, Bash, FastAPI, Flask, Spring Boot, REST APIs, GraphQL, Django
Frontend & Visualizations: React.js, Next.js, TypeScript, CSS, HTML
Databases & Data Engineering: RDS (PostgreSQL, MySQL), Vector Databases (FAISS, Pinecone), Snowflake, MongoDB, Redis, Apache Kafka, PySpark
Machine Learning Algorithms: Regression & Classification algorithms, Ensemble Learning (XGBoost, LightGBM), Time-Series Forecasting (LSTM, ARIMA), Anomaly Detection, Feature Engineering, Dimensionality Reduction (PCA), Hyperparameter Tuning, Model Evaluation, A/B Testing
NLP & Generative AI: LLM Development and Tuning, Transformers, BERT, RAG, Prompt Engineering, LangChain, Agentic Workflows, Embeddings, Semantic Search, Sentiment Analysis
ML Libraries & Frameworks: PyTorch, TensorFlow, Scikit-learn, Hugging Face Transformers, Statsmodels, Matplotlib, Seaborn
Cloud & MLOps: AWS (SageMaker, S3, EC2, Lambda), Docker, Kubernetes, CI/CD (GitHub Actions, Jenkins), ML Pipelines, Git, MLflow, Model Monitoring, Drift Detection, IaC (Terraform), Grafana, Prometheus

Experience

AI Software Engineer, State Street, USA

Sep 2025 – Present

- Spearheaded development of a RAG-based investment copilot using LangChain, OpenAI APIs, and FAISS, enhancing retrieval quality with re-ranking and cutting analyst research time by 50%.
- Developed predictive models with Scikit-learn and vector databases (FAISS) for anomaly detection in transaction data, incorporating feature engineering to reduce false alerts by 20% in early-stage projects.
- Implemented secure, low-latency APIs using FastAPI, GraphQL, JWT, and Redis caching, supporting high-concurrency financial queries while strengthening access control and response efficiency.
- Orchestrated event-driven data pipelines with Kafka and Celery, applying fault-tolerant streaming and partitioning strategies to accelerate multi-source data ingestion by 35%.
- Established LLMops workflows with LangChain and MLflow, incorporating prompt versioning, evaluation pipelines, and monitoring to improve response reliability and manage inference costs.

Machine Learning Engineer, Happiest Minds Technologies, INDIA

Apr 2021 – Dec 2023

- Engineered a predictive analytics platform for financial forecasting using Transformers and fine-tuning techniques with PyTorch on time-series data, achieving 22% better accuracy in market volatility prediction for investment clients.
- Collaborated with data science and DevOps teams to develop and tune churn prediction models using XGBoost, PySpark, and Optuna, improving recall-driven performance and reducing customer churn by 18% within six months.
- Architected an end-to-end document intelligence platform using BERT, AWS Textract, and Kafka, designing scalable pipelines that reduced manual processing by 65% and improved throughput by 3x.
- Built and optimized NLP extraction models with Python and FastAPI, applying feature engineering and validation logic to achieve 92%+ accuracy across diverse financial document formats.
- Designed containerized microservices with Docker, Kubernetes (EKS), and CI/CD pipelines, enabling automated model deployment, versioning, and reducing release cycles by 40%.

Education

Master in Computer Science, Clark University, Worcester, MA, USA

Jan 2024 – Dec 2025

Projects

Stock Market Trend Prediction Analysis with Deep Learning and ML (Link)

- Led a 4-member team to design and deploy a full-stack stock prediction web app using Python, Flask, HTML, and CSS, integrating dynamic data visualization for real-time market insights.
- Executed and compared ML and DL models (Linear Regression, LSTM, GRU) for stock forecasting, improving accuracy 25% and optimizing pipelines to cut latency 30%, ensuring production inference.

Credit Card Fraud Detection using Random Forest (Link)

- Engineered a scalable fraud detection system using Random Forest Classifier to flag anomalous transactions within a dataset of 984,000+ records.
- Enforced and compared ML and learning models, including Linear Regression, LSTM, and GRU, to forecast stock indices, improving accuracy 25% and optimizing pipelines to cut latency 30%.